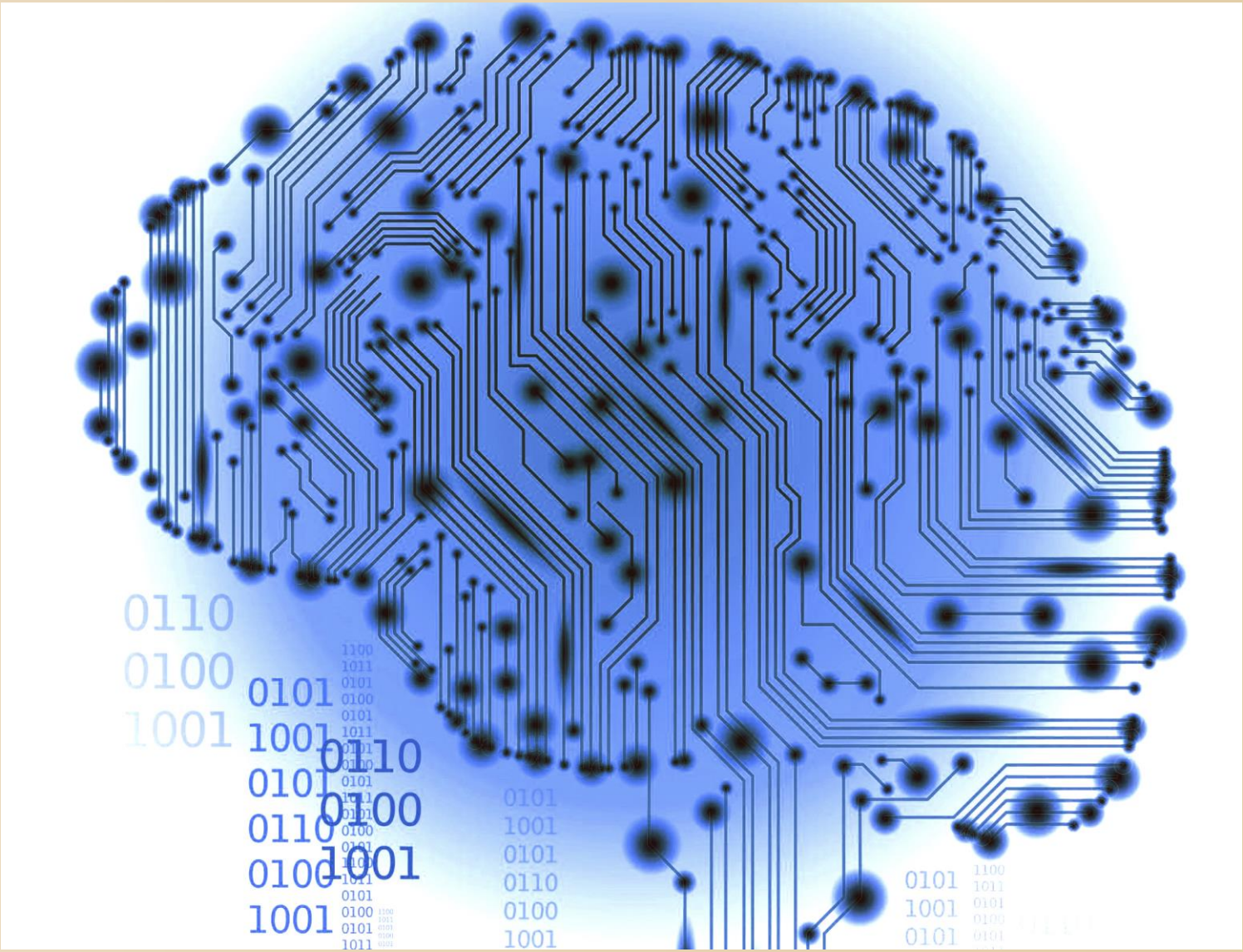
Computation



Before

Science is quantitative and seeks mechanisms that both explain and predict the behavior of the physical world. Therefore, calculation is very important to science.

Theories can be supported or invalidated by experimental data but this often requires extensive calculation.

The advent of computers, which became widely available in the last few decades of the 20th century, has provided scientists with computational capability that never existed before.

Counting

Counting preceded the invention of numbers and number systems. The oldest counting artifact is the Lebombo bone that dates to 35,000 BCE. The bone has 29 notches suggesting it was a calendar stick used for counting the days of the month.

Tally sticks are the first known written records. Counting, or in the commercial sense, accounting may have been the origin of writing.

Counting days and months is a way to predict seasons and determine when to store food, when to plant, and when to harvest.

Number Theory

Number systems were developed by several civilizations including the Egyptians, Indians, and Babylonians.

The decimal system we have today was an Arab-Indian invention of the 5th century CE.

As with so many things, the Ancient Greeks were the first to develop number theory.

Computation and computer are derived from computo in Latin. It means to count by a mathematical method

Abacus

The first known constructed counting device was the Chinese counting board which appeared about 400 BCE. This same device evolved over time and by about 1200 CE had become the abacus which is still in use today.

It was the introduction of the abacus to Europe that permitted the establishment of international banking during the Renaissance.

Abacus Tutorial: 1 Basic function

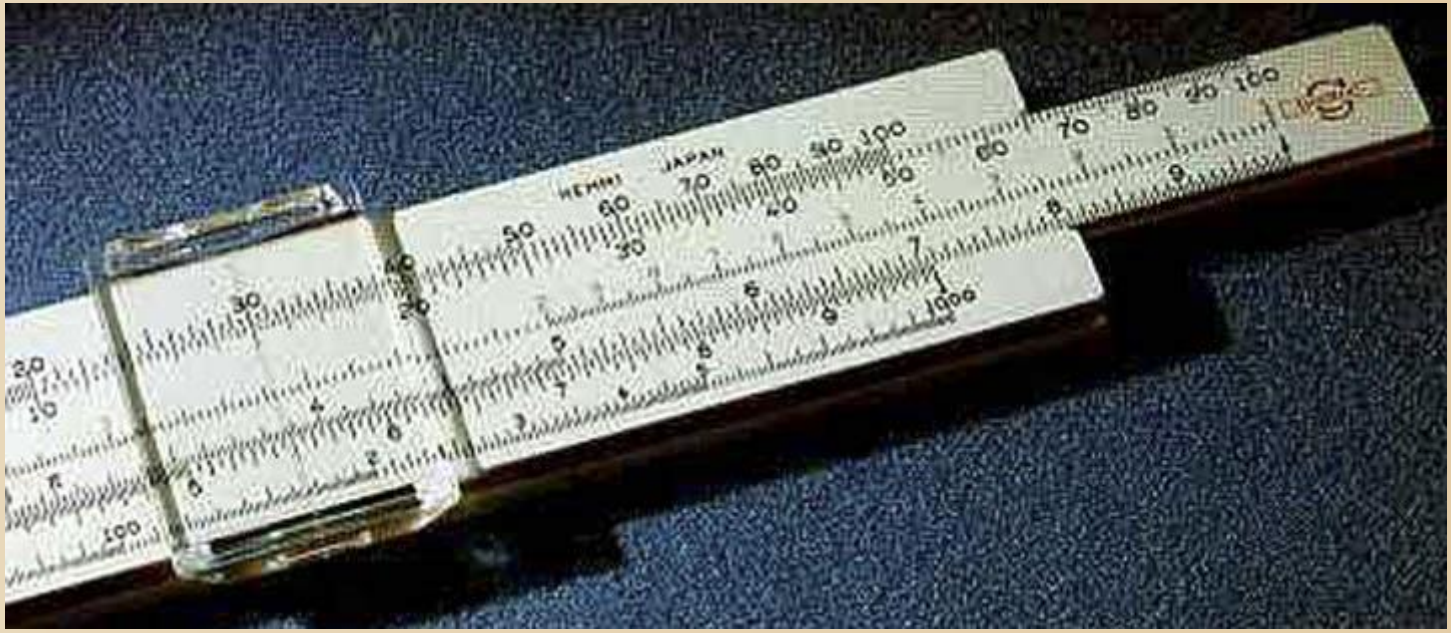
https://www.youtube.com/watch?v=FTVXUG_PngE

https://www.youtube.com/watch?v=I8EZvig5fOU

Slider Rules – since 1950s

For Engineering calculations:

https://www.youtube.com/watch?v=dT7bSn03lx0



From Mechanical Calculators to Computers

In 1642, Blaise Pascal, a French mathematician, built a mechanical adding machine that could add and subtract in 1693.

Leibniz increased the capability of mechanical calculators to multiply and divide.

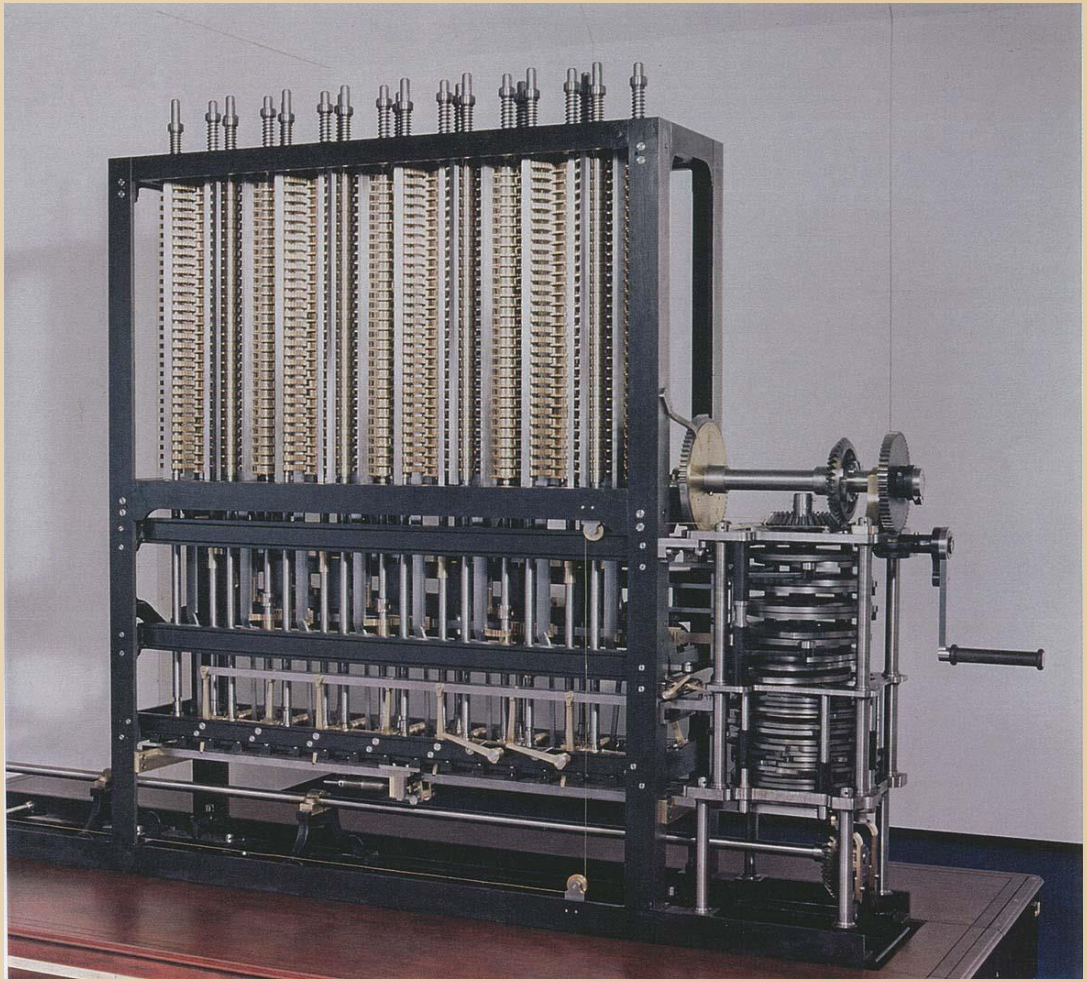
From Mechanical Calculators to Computers

The first person who attempted to move beyond the calculator to the computer was Charles Babbage (1791–1871).

Babbage wanted to make accurate mathematical tables which, to that point, had been produced by hand calculation, a process prone to error.

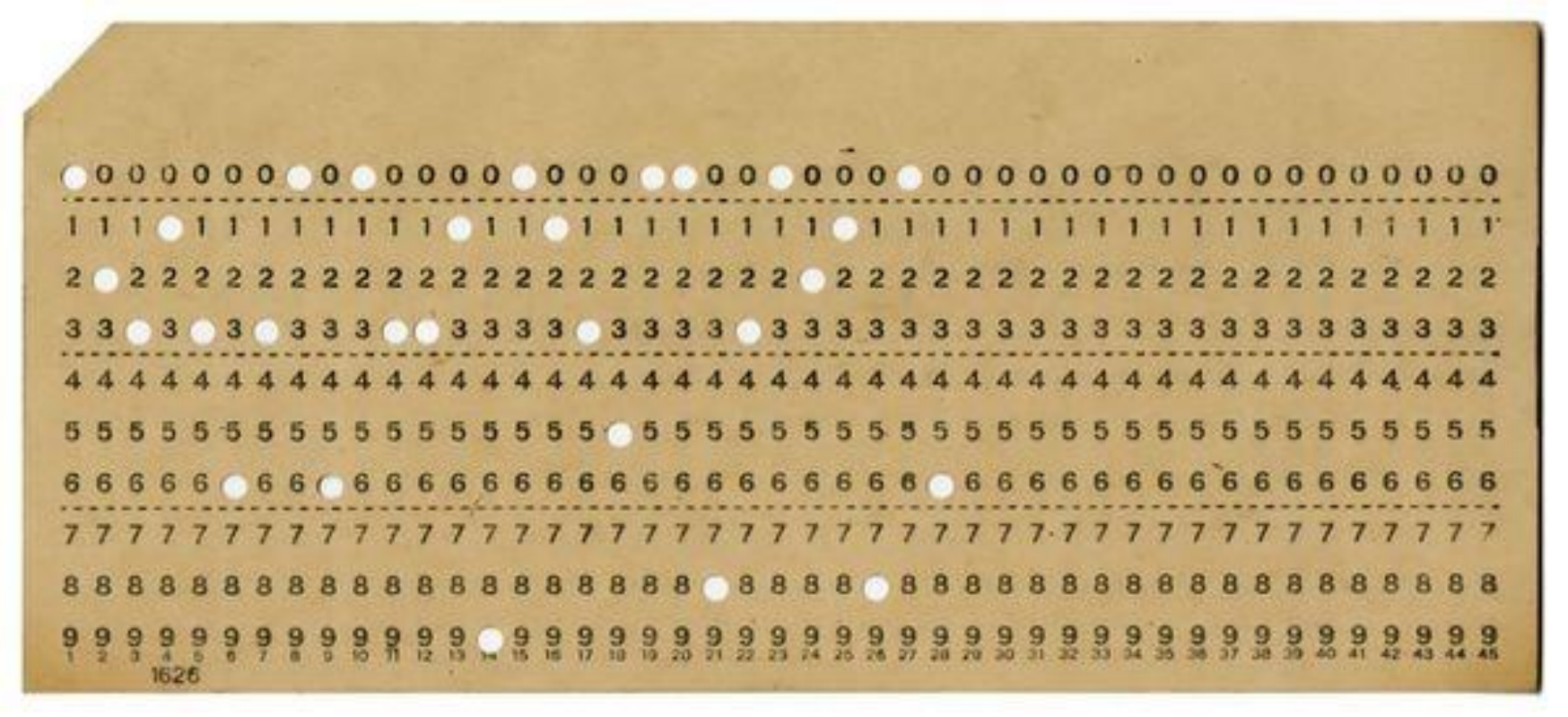
Early Computers

The Analytical Engine (by Babbage) incorporated an arithmetic logic unit, control flow in the form of conditional branching and loops, and integrated memory, making it the first design for a general-purpose computer.



Early Computers

The input, consisting of programs and data was to be provided to the machine via punched cards, a method being used at the time to direct mechanical looms such as the Jacquard loom.



Analog vs Digital

We need to differentiate between analog and digital computers. In general, digital means using numbers whereas analog means using some physical variable.

Digital devices are discrete while analog devices are continuous.

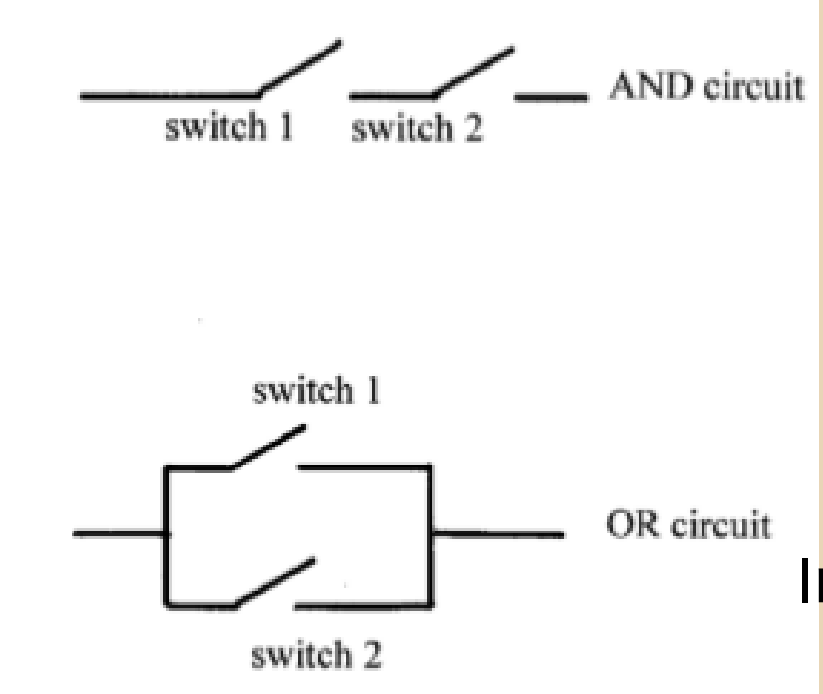
Analog signals are usually about 1% accurate. But digital signals can be made more accurate by adding more digits, virtually without limit.

Computing Machines

Claude Shannon (1916–2001), an American mathematician and electrical engineer, completed the step from pure mathematics to practical computing in his masters' thesis, A Symbolic Analysis of Relay and Switching Circuits at MIT in 1937.

Shannon brought Faraday and Boole together making the modern digital computer possible.

Computing Machines

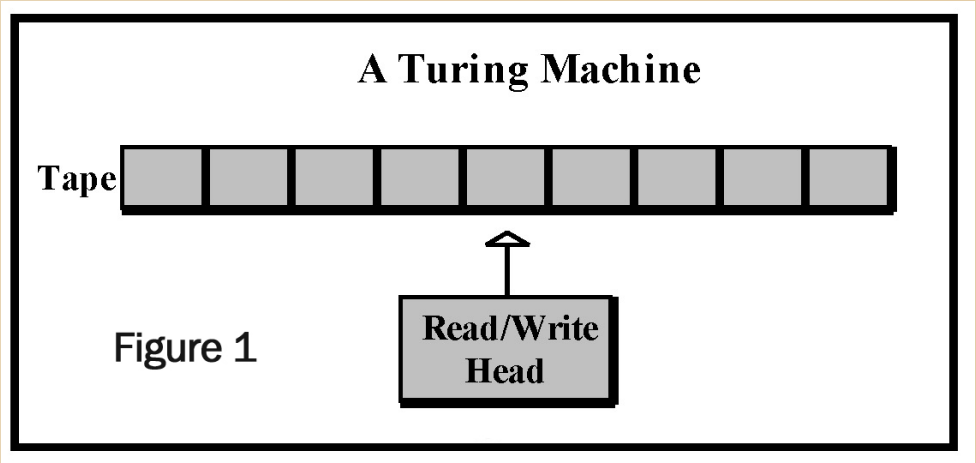


In the AND circuit two switches are in serial order. Thus, the circuit is complete only when both switches are closed. If we assign True to a closed switch, False to an open switch, True to a current flowing through the circuit and False to no current, then this circuit will give True when both Switches are True, and False at all other times.

In the OR circuit, the two switches are connected in parallel. Now the current flows (True response) whenever one or both of the switches are closed (True) and only gives a False response when both the switches are open (False).

Turing Machine - 1936

Alan Turing (1936) published a landmark paper on computing that defined a machine that would be capable of solving any conceivable mathematical problem for which an algorithm could be written.



Such a computer, now called a Turing Machine, would have internal states; internal operations, an infinitely long tape; and a read/write head.

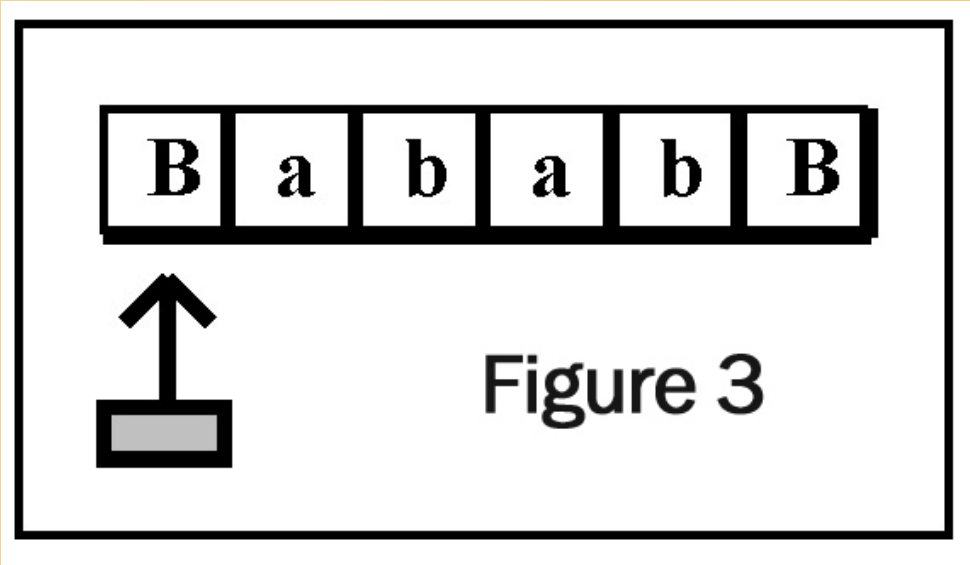
Alan Turning Machine

https://www.youtube.com/watch?v=gJQTFhkhwPA

Example Turing Machine

We'll set the starting position under the left blank. This is the initial state q0. What we want the Turing Machine to do is to change every a to b, and every b to a. We'll call this Turing Machine the Symbol Exchanger.

In order to program the Turing Machine to do the symbol exchange, we need to develop a set of rules, or transition functions, δ.



Assume we have a tape filled with the symbols a and b and framed by a blank symbol on each end, as shown in Figure 3.

Example Turing Machine

Here are the rules in English, then translated into Turing Machine symbols: Begin at the initial state, q0.

If a blank is over the read head, then re-copy a blank on the tape, set the state to the new state q1, and advance the read head to the right. Symbolically, we can write this as (q0 ,B) = {q1, B, R}. In general, this means (State, Symbol) = {NewState, NewSymbol, MoveTo}.

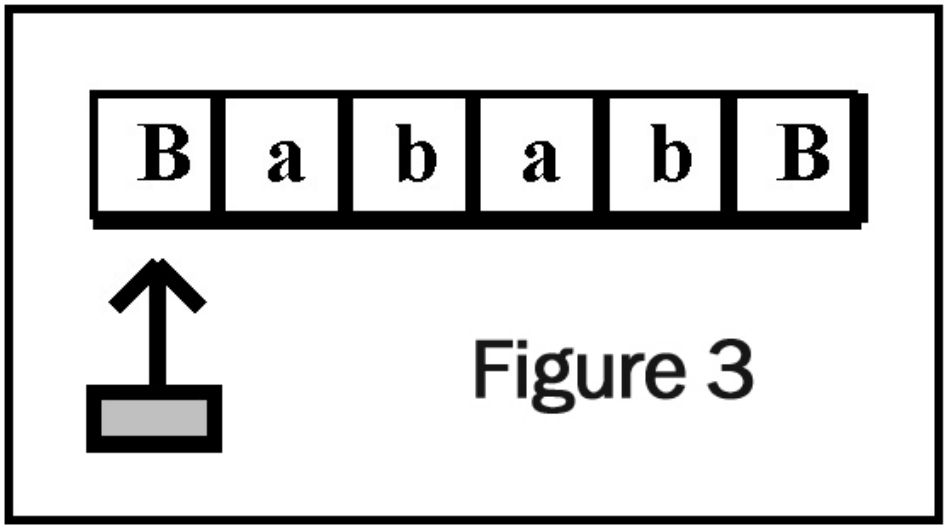


Figure 3

Assume we have a tape filled with the symbols a and b and framed by a blank symbol on each end, as shown in Figure 3.

Character Exchanger

Turing Machine for BababB

$$M = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$$

$$(q_0, B) = \{q_1, B, R\}$$
$$(q_1, a) = \{q_1, b, R\}$$
$$(q_1, b) = \{q_1, a, R\}$$
$$(q_1, B) = \{F\}$$

Figure 4

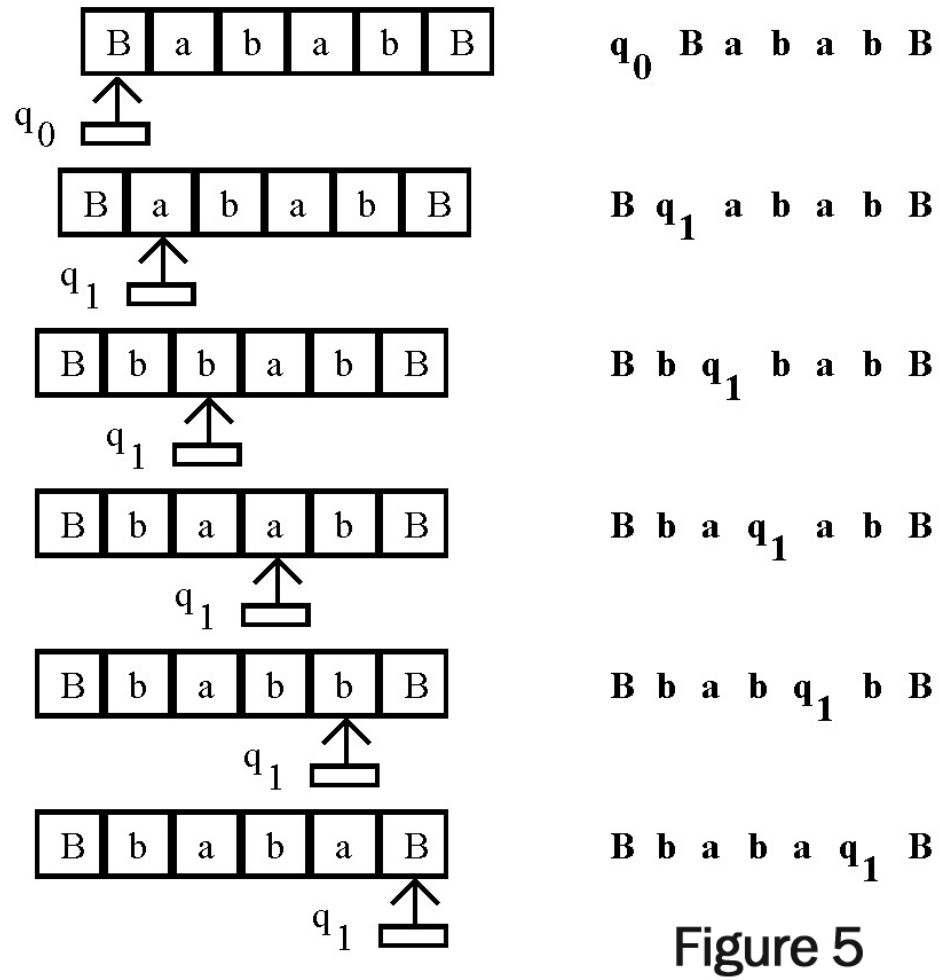


Figure 5

Adding 2 numbers–Turing Machine

0110 → 2

01110 → 3

2 + 2 → B0110110B

The result we want is four, or 011110. If we can shift the rightmost ones to the left one place, we'll have the desired result. What we need is a form of a shift register.

Adding 2 numbers–Turing Machine

Turing Machine to Add Two Integers

$$M = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$$

Input Tape = B0110110B

Transition Functions

$(q_0, 0) = \{q_1, 0, R\}$

$(q_1, 1) = \{q_1, 1, R\}$

$(q_1, 0) = \{q_2, 0, R\}$

$(q_2, 1) = \{q_3, 0, L\}$

$(q_2, 0) = \{q_5, B, R\} = \{F\}$

$(q_3, 0) = \{q_4, 1, R\}$

$(q_4, 1) = \{q_1, 0, L\}$

$(q_4, 0) = \{q_2, 0, R\}$

Computation Trace

$q_0\ 0110110 \vdash 0q_1 110110 \vdash 01\, q_1 10110 \vdash 011\, q_1 0110$

$\vdash 0110\ q_2 110 \vdash 011\ q_3 0010 \vdash 0111\ q_4 010 \vdash 01110\ q_2 10$

$\vdash 0111\ q_3 000 \vdash 01111\ q_4 00 \vdash 011110\ q_2 0 \vdash 011110B\ q_5 \vdash \{F\}$

Figure 6

Von Neumann 1945

John von Neumann (1903–1957) was born in Budapest. In 1930 Von Neumann became one of the first four faculty members at the Advanced Institute at Princeton.

- Einstein and Gödel were also in this group.

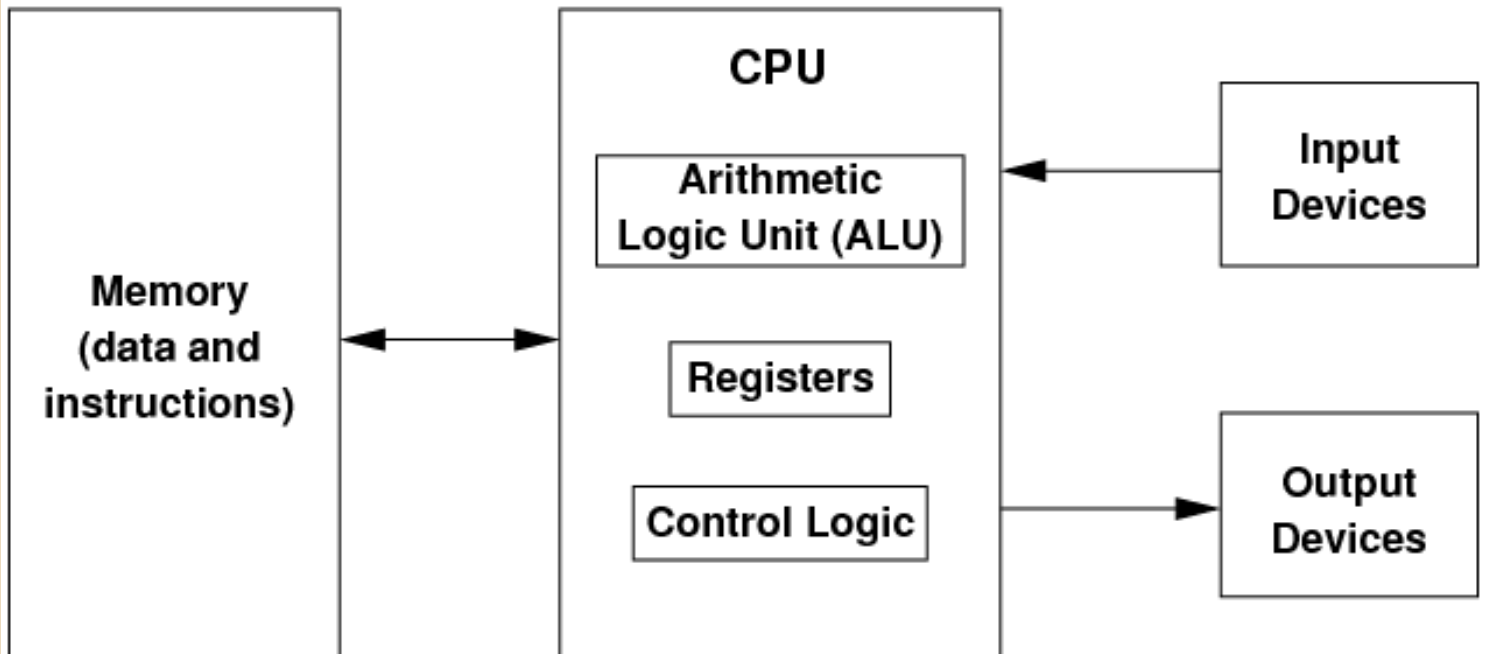
He participated in the Manhattan project and helped design the atomic bomb.

- It is possible that the brain cancer that killed him in 1957 was induced by his observation of the test of the atomic bomb at Trinity Site on July 16, 1945

Von Neumann 1945

Von Neumann's greatest contribution to the development of computers was designing a flexible architecture so that both instructions and data could be stored in single storage area.

The von Neumann architecture is the basic design for all modern computers.



Turing Machine - 1936

During WW II, he worked on the English effort in cryptography and helped break the German Enigma Code.

Turing built an electromechanical machine that speeded up the breaking of coded message. Breaking the German code gave the British and Americans the ability to know where the German submarines are stationed.

https://www.youtube.com/watch?v=G2_Q9FoD-oQ

Mathematics of Computers
Computer Science

George Boole (1815-1864)

Boole claimed that logic was really a field of mathematics, not philosophy as it had been considered since Aristotle invented it.

Boole turned Aristotle's formal logic into a mathematical form where symbols could be manipulated.

By representing TRUE as 1 and FALSE as 0, the operations AND, OR, and NOT, could be performed mathematically. However, this requires new sets of rules.

Boolean Algebra

For the AND operator:

1 + 1 = 1	TRUE AND TRUE = TRUE
1 + 0 = 0	TRUE AND FALSE = FALSE
0 + 1 = 0	FALSE AND TRUE = FALSE
0 + 0 = 0	FALSE AND FALSE = FALSE

But, for the OR operator:

1 + 1 = 1	TRUE OR TRUE = TRUE
1 + 0 = 1	TRUE OR FALSE = TRUE
0 + 1 = 1	FALSE OR TRUE = TRUE
0 + 0 = 0	FALSE OR FALSE = FALSE

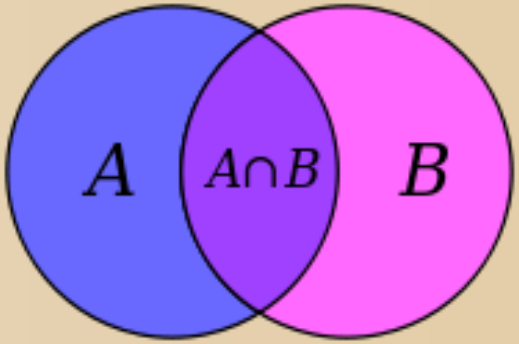
Decimal	Binary
0	0
1	1
2	10
3	11
4	100
5	101
6	110
7	111
8	1000
9	1001
10	1010
11	1011
12	1100
13	1101
etc.	

Two binary numbers can be added, in fact, more simply than decimal numbers. If both are 0, then the sum is 0; if one is 1 and the other 0, then the sum is 1; and, if both are 1 then the sum is 0 but a 1 is carried to the left.

Set Theory

George Cantor developed Set Theory in 1847 based on Boolean algebra.

Set theory as a foundation for mathematical analysis, topology, abstract algebra, and discrete mathematics



Set theory begins with a fundamental binary relation between an object o and a set A. If o is a member of A, the notation o ∈ A is used.

Since sets are objects, the membership relation can relate sets as well.

Completeness vs Incompleteness

During the late 19 th and early 20 th centuries, mathematicians made progress on defining formal systems of mathematics with the goal of finding a system that is complete.

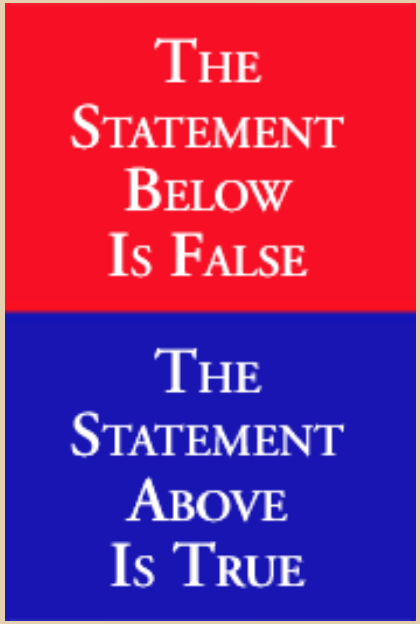
By complete, we mean a system of mathematics that can solve all problems, prove all true theorems and disprove all false theorems.

Incompleteness Theorem (1931)

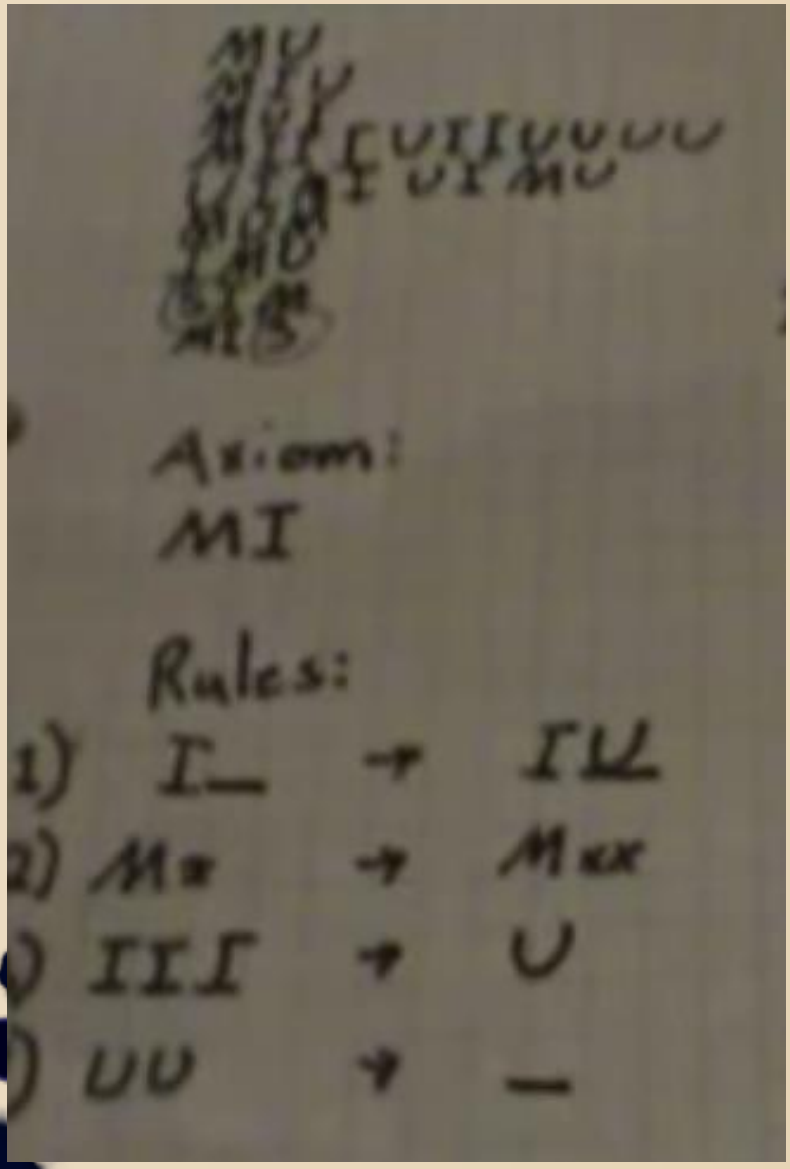
Kurt Gödel eliminated the possibility that any system of mathematics could be complete.

Gödel proved that in any system there would be theorems that were true but could never be proven and theorems that were false but could never be disproven.

An easy example of the Incompleteness Theorem to understand is called The Liar's Paradox. If someone said to you: "I am lying." would you believe them?



Example Incompleteness



Start with single M in the MIU System: MIU, MU well formed.

Axiom: MI

R2: MII

R2: MIIII

R2: MIIIIIIII

R3: MIUUI

R4: MII

R1: MIIU

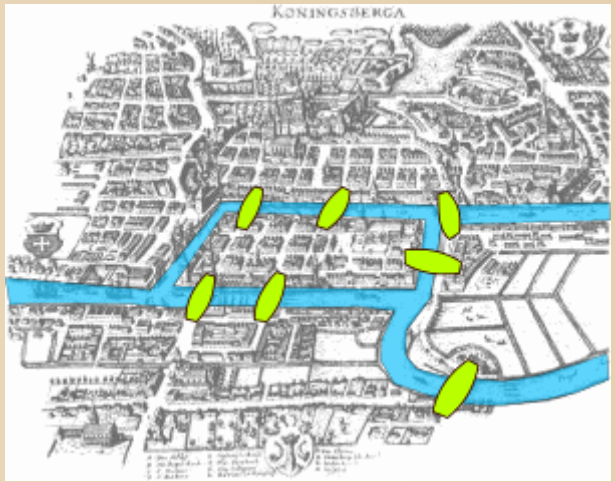
Is MU a theorem ?

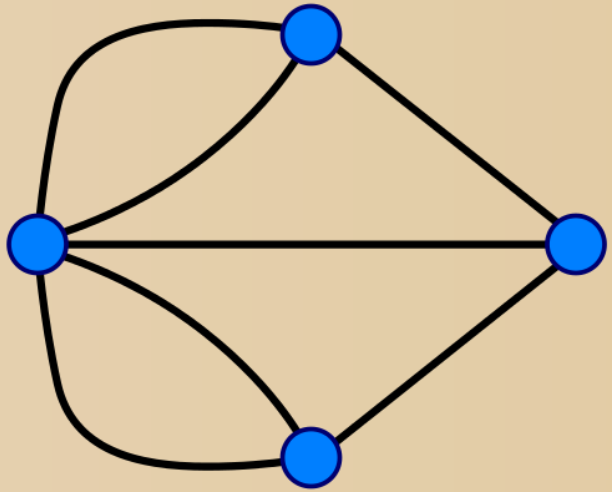
Can you prove if it exist or not exists ?

Euler

Swiss mathematician Leonhard Euler was probabily the most important mathematicians of all time.

- The irrational mathematical constant e (2.71828...) is named after Euler.
- He approached many problems of engineering analytically. - Eulars Methods
- Graph Theory Presented solution to "Seven Bridges of Königsberg"





NP Complete

The concept of NP-completeness was introduced in 1971 (see Cook–Levin theorem), though the term NP-complete was introduced later.

Although any given solution to an NP-complete problem can be verified quickly (in polynomial time), there is no known efficient way to locate a solution in the first place; indeed, the most notable characteristic of NP-complete problems is that no fast solution to them is known.

NP Complete

That is, the time required to solve the problem using any currently known algorithm increases very quickly as the size of the problem grows.

While a method for computing the solutions to NP-complete problems using a reasonable amount of time remains undiscovered, computer scientists and programmers still frequently encounter NP-complete problems. NP-complete problems are often addressed by using heuristic methods and approximation algorithms.

Computer Documentary

https://www.youtube.com/watch?v=D7Y2C_rFvo0